## CoM standard module classes

COM-HPC modules represent the next step in embedded systems, providing advanced features to meet the increasing need for high-speed, scalable performance. They come in three types, each suited to different needs, from powerful computing in harsh conditions to compact, space-saving solutions. The table on first page compares the key features of these three COM-HPC module classes.

It is important to note that power capacity for all COM-HPC modules can vary based on factors such as connector pin de-rating (which typically reduces the rated power by 20%), the number of memory sockets utilised, and the power demands of other components on the module. These variables must be taken into account when designing a system to ensure optimal performance and longevity.

## COM-HPC Mini simplifies embedded computing

The COM-HPC Mini targets applications requiring compact, powerful computing solutions, including mobile systems in medical technology, rugged battery-powered tablets, and size-constrained automated guided vehicles (AGVs) and autonomous mobile robots (AMRs). It is likely to become a preferred choice in these areas.

### COM-HPC Mini vs COM-HPC

- ***Connector configuration.*** COM-HPC Mini utilises a single high-speed connector, whereas the larger COM-HPC Client and Server use two connectors
- ***Compatibility.*** The design changes in the COM-HPC Mini, including the connector and interface adjustments, result in incompatibility with the Client and Server
- ***Signal pin capacity.*** Although COM-HPC Mini has half the number of signal pins compared to larger modules, it still provides 400 high-speed signal lanes, representing 90% of the capacity of COM Express Type 6 modules
- ***Memory configuration.*** COM-HPC Mini modules require soldered memory to enhance ruggedness, improving resistance to shock and vibration, and increasing overall efficiency
- ***Market position.*** Unlike other small form factor standards like COM Express Mini, COM-HPC Mini is designed as a complementary standard that meets the needs of applications requiring high performance in a compact space
- ***Size and application.*** COM-HPC Mini is significantly smaller than the next-smallest COM-HPC form factor and is specifically targeted for cost-effective, lower-power applications in autonomous mobile robots, drones, mobile 5G test and measurement equipment, and other far-edge applications

The COM-HPC Mini complements existing standards like COM Express Mini to meet new demands, suitable for developers upgrading or creating designs without needing the highest performance or advanced interface technology. Many designs can operate effectively without maximum performance and bandwidth.

Credit-card-sized modules have shown the need for diverse standards that coexist, fulfilling various roles over decades. The COM-HPC Mini uses a single connector, unlike the two in the larger COM-HPC Client and Server modules. With 400 signal pins, it delivers much of the performance capacity of the COM Express Type 6 modules but in a smaller size. This size fits embedded computing in specialised devices, such as top-hat rail PCs in control cabinets for building and industrial automation, and portable test and measurement equipment.

The specification supports PCIe Gen4 and Gen5, ensuring performance remains intact. Despite a reduction in pin count, the COM-HPC Mini seamlessly integrates into the broader COM-HPC ecosystem.

The COM-HPC Mini includes I/O capabilities such as USB4 and SoundWire, along with Ethernet capabilities through 2x NBaseT and 2x RGMII ports shared with PCIe lanes. It supports the SATA interface, but newer applications are expected to use the NVMe interface through dedicated PCIe lanes. The design optimises pin usage by sharing 8 high-speed data lanes among various interfaces, including DDI (digital display interface), USB 3.x, and USB4.

### The features at a glance

- ***Compact size.*** Measuring only 95mm x 70mm, significantly smaller than the COM-HPC Client Size A, ideal for space-limited applications
- ***Interface adjustment.*** Includes a shift to a single high-speed connector with a distinct pinout to minimise form factor and enhance functionality, supporting interfaces like USB 4.0, Thunderbolt, PCIe Gen4/5, and 10Gbit/s Ethernet
- ***Incompatibility with larger modules.*** Design adjustments make COM-HPC Mini modules incompatible with the client and server specifications
- ***High performance capacity.*** Despite its compact form, it supports up to 76 watts of power consumption, enabling high-performance processors like Intel Core
- ***Reduced heat dissipation.*** Specifies a reduced maximum heat dissipation in line with its compact and power-efficient design
- ***Rugged design.*** The overall height of the module and heat spreader is reduced by 5mm, requiring only 15mm above the carrier surface. This reduction facilitates integration into slim designs like mobile devices and panel PCs
- ***Soldered memory.*** Incorporates soldered memory to enhance durability and resistance to shock and vibrations, and improves cooling efficiency through direct thermal coupling to the heat spreader

## The strategic shift to COM-HPC Mini

The physical build of the COM-HPC Mini further underscores its utility in demanding environments. The move to a lower profile has led to the adoption of soldered memory, which is less susceptible to damage from shock and vibrations and allows for direct thermal coupling to heat spreaders, enhancing the module's durability and thermal efficiency. These features ensure the COM-HPC Mini's performance remains